

comparative_analysis

January 20, 2026

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[8]: # =====
# CLASSICAL vs DEEP LEARNING - UNIFIED COMPARATIVE ANALYSIS
# (DISPLAY ONLY - NO FILE SAVING)
# =====
# PURPOSE:
# - Compare Classical ML and Deep Learning models fairly
# - Use identical core metrics
# - Provide performance-efficiency trade-off analysis
# - Display tables, plots, and detailed interpretations
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from IPython.display import display, Markdown

# -----
# 1 PATH CONFIGURATION
# -----
# Assumption:
# - This script is executed from /code
# - Results directory exists at project root /results
# -----

CODE_DIR = Path.cwd()
PROJECT_ROOT = CODE_DIR.parent

CLASSICAL_CSV = PROJECT_ROOT / "results/classical_comparison.csv"
DEEP_CSV = PROJECT_ROOT / "results/deep_learning_comparison_metrics.csv"

# -----
# 2 CORE METRIC DEFINITION
# -----
# These columns MUST exist and MUST be in the same order
# for both Classical and Deep Learning CSVs
# -----
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CORE_COLUMNS = [
    "Model",
    "Accuracy",
    "Precision (Macro)",
    "Recall (Macro)",
    "F1-score (Macro)",
    "Training Time (s)",
    "Inference Time / Sample (s)",
]

# Metrics where higher values indicate better performance
HIGHER_IS_BETTER = [
    "Accuracy",
    "Precision (Macro)",
    "Recall (Macro)",
    "F1-score (Macro)",
]

# Metrics where lower values indicate better efficiency
LOWER_IS_BETTER = [
    "Training Time (s)",
    "Inference Time / Sample (s)",
]

# -----
# 3 LOAD CSV FILES
# -----

classical_df = pd.read_csv(CLASSICAL_CSV)
deep_df = pd.read_csv(DEEP_CSV)

# -----
# 4 VALIDATION FUNCTION
# -----
# Ensures:
# - All core columns exist
# - Core columns appear in the SAME ORDER
# - Extra columns (Ranks etc.) are allowed
# -----

def validate_core_columns(df, name):
    missing = [c for c in CORE_COLUMNS if c not in df.columns]
    if missing:
        raise ValueError(f" Missing columns in {name}: {missing}")

    core_order = [c for c in df.columns if c in CORE_COLUMNS]

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    if core_order != CORE_COLUMNS:
        raise ValueError(
            f" Core column order mismatch in {name}\n"
            f"Expected: {CORE_COLUMNS}\n"
            f"Found    : {core_order}"
        )

    print(f" {name} validated")

validate_core_columns(classical_df, "Classical ML CSV")
validate_core_columns(deep_df, "Deep Learning CSV")

# -----
# 5 UNIFIED BENCHMARK TABLE
# -----
# Combines Classical + Deep Learning metrics
# for direct comparison
# -----

benchmark_df = pd.concat(
    [
        classical_df[CORE_COLUMNS],
        deep_df[CORE_COLUMNS],
    ],
    ignore_index=True
)

display(Markdown("## Unified Benchmark Metrics"))
display(
    benchmark_df.style
        .format("{:.4f}", subset=CORE_COLUMNS[1:])
        .set_properties(**{"text-align": "center"})
)

# -----
# 6 NORMALIZED SCORE INDEX (NSI)
# -----
# NSI captures:
# - Classification quality
# - Computational efficiency
# All metrics are normalized to [0, 1]
# Higher NSI = better overall trade-off
# -----

nsi_df = benchmark_df.copy()

# Normalize higher-is-better metrics

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for col in HIGHER_IS_BETTER:
    min_v, max_v = nsi_df[col].min(), nsi_df[col].max()
    nsi_df[col + "_norm"] = (nsi_df[col] - min_v) / (max_v - min_v + 1e-9)

# Normalize lower-is-better metrics
for col in LOWER_IS_BETTER:
    min_v, max_v = nsi_df[col].min(), nsi_df[col].max()
    nsi_df[col + "_norm"] = (max_v - nsi_df[col]) / (max_v - min_v + 1e-9)

# Final NSI score
norm_cols = [c for c in nsi_df.columns if c.endswith("_norm")]
nsi_df["NSI"] = nsi_df[norm_cols].mean(axis=1)

nsi_df = nsi_df.sort_values("NSI", ascending=False)

display(Markdown("## Normalized Score Index (NSI)"))
display(
    nsi_df[["Model", "NSI"]]
    .style.format({"NSI": "{:.3f}"})
    .bar(subset=["NSI"], color="#4CAF50")
)

display(Markdown(
    """
    ### What NSI Indicates
    - NSI combines accuracy and efficiency
    - All metrics are normalized to [0, 1]
    - Higher NSI better accuracy-speed trade-off
    """))

# -----
# 7 PLOT 1: ACCURACY vs TRAINING TIME (SCATTER PLOT)
# -----
# Encodes:
# - X-axis: Training Time
# - Y-axis: Accuracy
# - Color: Model type (Classical vs Deep Learning)
# - Optional: Inference Time via marker size (scaled)
# -----

display(Markdown("## Accuracy vs Training Time (Scatter Plot)"))

plt.figure(figsize=(8, 5))

# Assign model type

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benchmark_df["Model Type"] = ["Classical ML"] * len(classical_df) + ["Deep Learning"] * len(deep_df)

# Optional: scale marker size by inference time
marker_sizes = benchmark_df["Inference Time / Sample (s)"] * 2000 # scale factor

colors = {"Classical ML": "#1f77b4", "Deep Learning": "#ff7f0e"}

for mtype in benchmark_df["Model Type"].unique():
    subset = benchmark_df[benchmark_df["Model Type"] == mtype]
    plt.scatter(
        subset["Training Time (s)"],
        subset["Accuracy"],
        s=marker_sizes[subset.index],
        c=colors[mtype],
        label=mtype,
        alpha=0.7,
        edgecolors="k"
    )

# Annotate models
for _, r in benchmark_df.iterrows():
    plt.text(
        r["Training Time (s)"] * 1.01, # slight shift to avoid overlap
        r["Accuracy"],
        r["Model"],
        fontsize=9
    )

plt.xlabel("Training Time (s)")
plt.ylabel("Accuracy")
plt.title("Accuracy vs Training Time\n(Color = Model Type, Marker size proportional to Inference Time)")
plt.legend(title="Model Type")
plt.grid(True)
plt.show()

# -----
# 8 PLOT 2: INFERENCE TIME COMPARISON (HORIZONTAL BAR)
# -----
# Focuses on:
# - Latency ranking
# - Accuracy shown as annotation
# Ideal for real-time deployment discussion
# -----

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display(Markdown("## Inference Time per Sample (Latency Comparison)"))

sorted_df = benchmark_df.sort_values(
    "Inference Time / Sample (s)",
    ascending=True
)

plt.figure(figsize=(8, 5))
plt.barh(
    sorted_df["Model"],
    sorted_df["Inference Time / Sample (s)"]
)

for i, (_, r) in enumerate(sorted_df.iterrows()):
    plt.text(
        r["Inference Time / Sample (s)"],
        i,
        f"Acc={r['Accuracy']:.3f}",
        va="center",
        fontsize=9
    )

plt.xlabel("Inference Time / Sample (s)")
plt.title("Inference Latency Comparison\n(Annotated with Accuracy)")
plt.grid(axis="x")
plt.show()

# -----
# 9 OVERALL NSI RANKING PLOT
# -----

display(Markdown("## Overall Model Ranking (NSI)"))

plt.figure(figsize=(8, 5))
plt.barh(nsi_df["Model"], nsi_df["NSI"])
plt.xlabel("Normalized Score Index (NSI)")
plt.gca().invert_yaxis()
plt.grid(axis="x")
plt.show()

# -----
# FINAL TEXTUAL ANALYSIS
# -----

best = nsi_df.iloc[0]

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display(Markdown(
f"""
## Comparative Analysis Summary

### Best Overall Model
- {best['Model']}
- NSI Score: {best['NSI']:.3f}

### Key Observations
- Deep learning models achieve high accuracy but incur significant training and inference costs
- Classical ML models remain computationally efficient and are better suited for real-time or edge deployment

### Final Conclusion
Accuracy alone is insufficient for fair comparison.
The Normalized Score Index (NSI) provides a principled,
quantitative way to balance performance and efficiency
across fundamentally different learning paradigms.
"""
))

print(" Comparative analysis displayed successfully")

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Classical ML CSV validated
Deep Learning CSV validated

0.1 Unified Benchmark Metrics

<pandas.io.formats.style.Styler at 0x772ec2b6ca50>

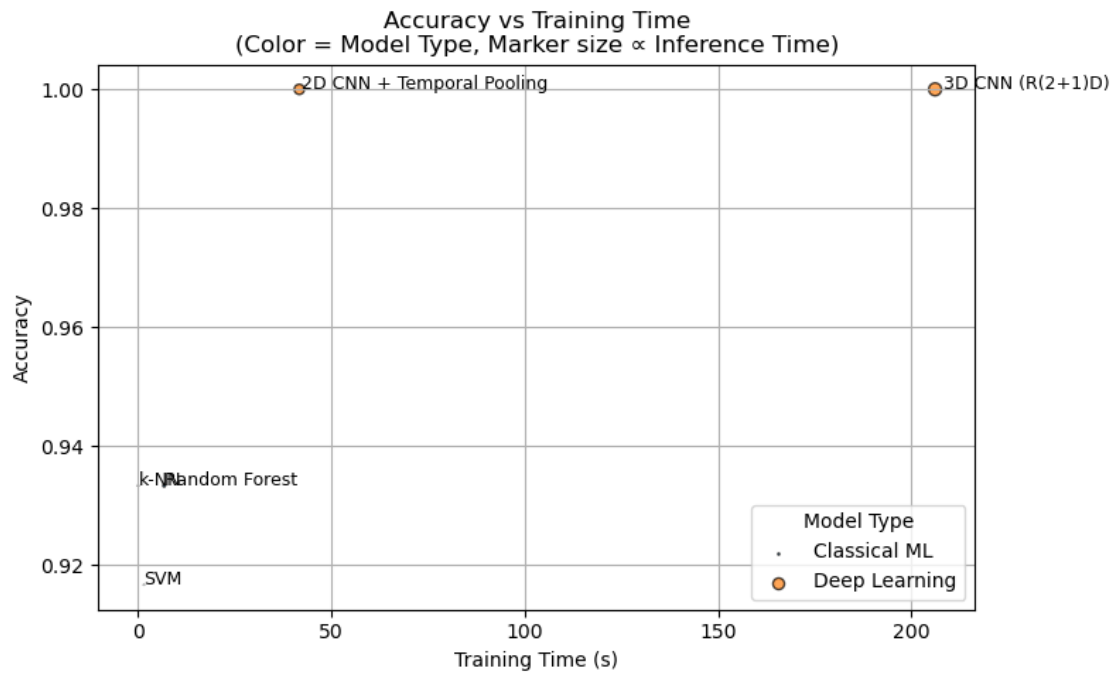
0.2 Normalized Score Index (NSI)

<pandas.io.formats.style.Styler at 0x772eedd289d0>

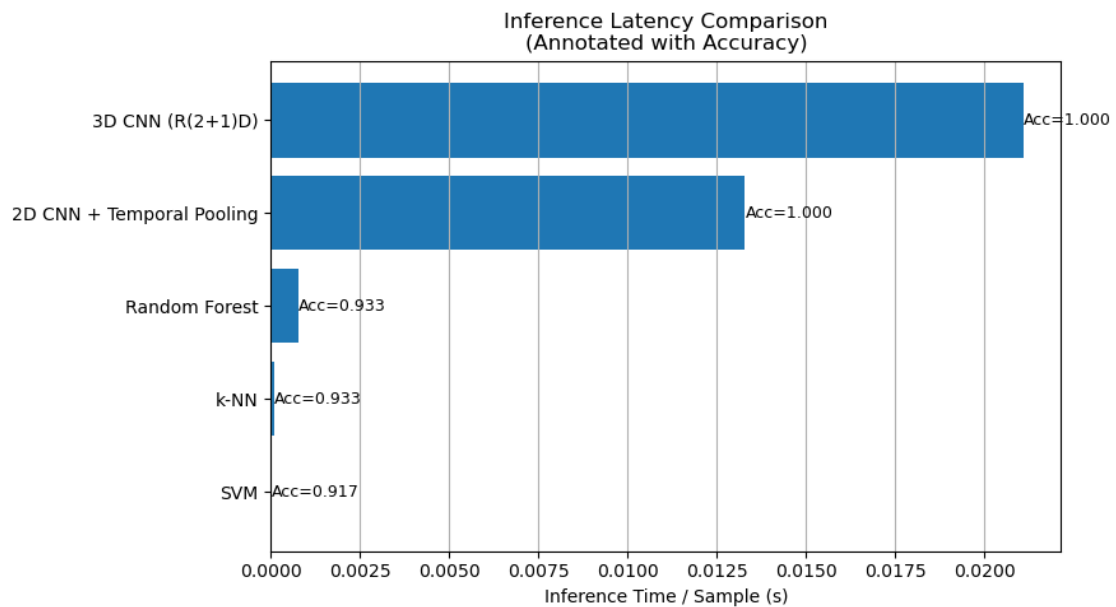
0.2.1 What NSI Indicates

- NSI combines **accuracy and efficiency**
- All metrics are normalized to **[0, 1]**
- Higher NSI = better accuracy-speed trade-off

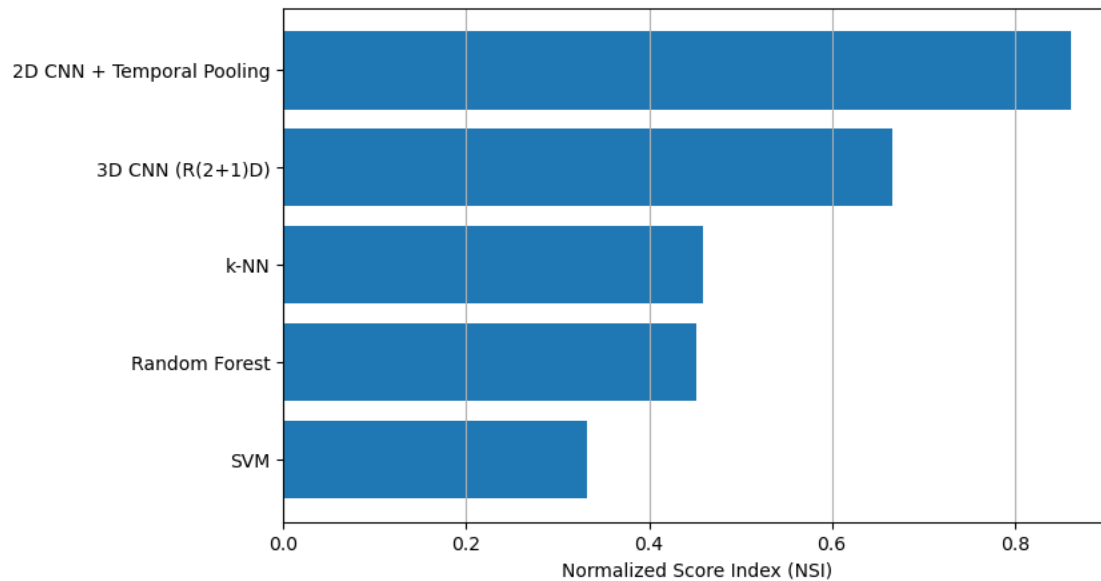
0.3 Accuracy vs Training Time (Scatter Plot)



0.4 Inference Time per Sample (Latency Comparison)



0.5 Overall Model Ranking (NSI)



0.6 Comparative Analysis Summary

0.6.1 Best Overall Model

- 2D CNN + Temporal Pooling
- NSI Score: 0.861

0.6.2 Key Observations

- Deep learning models achieve **high accuracy** but incur **significant training and inference costs**
- Classical ML models remain **computationally efficient** and are better suited for **real-time or edge deployment**

0.6.3 Final Conclusion

Accuracy alone is insufficient for fair comparison. The **Normalized Score Index (NSI)** provides a principled, quantitative way to balance **performance and efficiency** across fundamentally different learning paradigms.

Comparative analysis displayed successfully

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